

Predictive Emergence of Hallucination-Conditioned Internal Representations in a Multimodal LLM

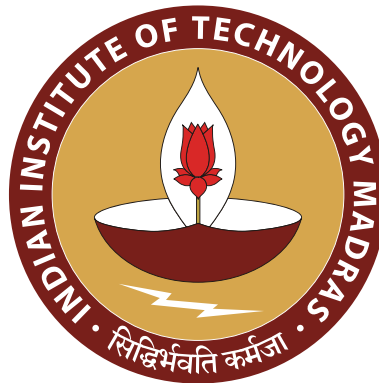
Submitted By

Basavaraj A Naduvnamani

Registration No: DA25C005

Under the Guidance of

Dr. Mitesh Khapra



**Department of Data Science and Artificial
Intelligence**

DA5410 Winter Project — Industrial Placement

25 NOV 2025 — 03 JAN 2026

DATA SHEET

DEPARTMENT OF DATA SCIENCE AND ARTIFICIAL INTELLIGENCE
INDIAN INSTITUTE OF TECHNOLOGY MADRAS

Name of the Student : Basavaraj A Naduvinamani
Registration No : DA25C005
Project Title : Predictive Emergence of Hallucination-Conditioned
Internal Representations in a Multimodal LLM
Course / Program : DA5410 Winter Project — Industrial Placement
Department : Department of Data Science and Artificial Intelligence
Name of the Supervisor : **Dr. Mitesh Khapra**
Project Duration : 25 November 2025 – 03 January 2026

Basavaraj A Naduvinamani
Student

Dr. Mitesh Khapra
Supervisor

Acknowledgement

I would like to express my sincere gratitude to **Dr. Mitesh Khapra** for his guidance and support throughout the DA5410 Winter Project. His academic direction and feedback encouraged a structured and research-oriented approach to investigating hallucination in multimodal language models.

I am deeply thankful to **Mohammed Safi Ur Rahman Khan** for his valuable mentorship during the course of this project. His technical insights, discussions, and constructive suggestions were instrumental in refining the experimental design and strengthening the analytical framework of this study.

I also acknowledge the AI4Bharat Lab at IIT Madras for providing a collaborative research environment and the necessary intellectual and computational resources that enabled this work.

Finally, I am grateful to the Department of Data Science and Artificial Intelligence at IIT Madras for fostering an academic setting that supports independent exploration and research-driven learning.

This project has significantly strengthened my interest in representation-level analysis and the internal dynamics of multimodal AI systems, and it has motivated me to continue exploring this research direction further.

Abstract

Hallucination in multimodal large language models (LLMs) remains a critical challenge, particularly in settings where visual and textual reasoning interact. While prior work has focused on output-level detection and mitigation through prompt engineering or post-hoc filtering, comparatively less attention has been directed toward understanding the internal representational dynamics preceding hallucinated responses.

In this work, we investigate whether hallucination-conditioned internal states in a multimodal transformer exhibit structured geometric separability prior to final token generation. Using LLaVA as a case study, we construct contrastive truth–hallucination prompt pairs and extract layer-wise hidden representations across the decoder stack. Using principal component analysis and layer-wise logistic regression, we evaluate the linear separability of hallucination-conditioned states.

Our findings reveal a consistent emergence window between Layers 13–17, where predictive accuracy surpasses 90%. This separability remains stable under random-direction controls, random-label baselines, bootstrap resampling, and cross-task validation, indicating that the observed structure is not an artifact of dataset construction or classifier bias.

These results suggest that hallucination corresponds to internally consolidating representational states that emerge prior to explicit token commitment. By reframing hallucination as an internal geometric phenomenon rather than a purely output-level anomaly, this study establishes a principled and extensible foundation for future large-scale and multi-model investigations into predictive internal dynamics in multimodal architectures.

Contents

1	Introduction	1
2	Related Work	2
3	Problem Formulation	3
3.1	Representational Separability	3
3.2	Emergence Layer	3
3.3	Geometric Distance Measure	3
3.4	Predictive Internal Commitment Hypothesis	4
4	Experimental Setup	4
4.1	Model Selection	4
4.2	Dataset Construction	4
4.3	Hidden State Extraction	4
4.4	Dimensionality Reduction and Axis Extraction	5
4.5	Classification Framework	5
4.6	Geometric Distance Analysis	5
4.7	Token Probability Dynamics (Exploratory)	5
4.8	Validation Controls	5
4.9	Overall Experimental Pipeline	6
4.10	Projection onto Truth Axis	7
4.11	Layer-wise Prediction Accuracy	8
4.12	Geometric Distance Analysis	10
4.13	Token Probability Dynamics	11
5	Results	11
5.1	PCA-Based Separability	11
5.2	Layer-wise Prediction Accuracy	12
5.3	Emergence Threshold Analysis	12
5.4	Bootstrap Confidence Estimation	12
5.5	Geometric Distance Dynamics	12
5.6	Token Probability Dynamics (Exploratory)	13
5.7	Validation Controls	13
5.8	Summary of Findings	13
6	Analysis and Interpretation	14
6.1	Internal Consolidation of Hallucination	14
6.2	Representational Pressure Accumulation	14
6.3	Predictive Separability and Early Detection Potential	14
6.4	Relation to Steering Experiments	14
6.5	Broader Implications	14
7	Limitations	14
7.1	Dataset Scale	14
7.2	Model Scope	14
7.3	Modal Coverage	15
7.4	Causal Intervention	15
8	Future Work	15
8.1	Large-Scale Dataset Expansion	15

8.2	Multi-Model Comparative Study	15
8.3	Temporal and Multimodal Extension	15
8.4	Early-Warning Hallucination Detection	15
8.5	Structured Causal Intervention	15
9	Research Outlook	15
10	Conclusion	16

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in language understanding, reasoning, and generation tasks. The recent emergence of multimodal architectures has extended these capabilities by integrating visual inputs alongside text, enabling models to perform image-grounded reasoning, visual question answering, and multimodal dialogue [1], [2]. Despite these advances, hallucination remains a persistent and critical challenge. Multimodal LLMs often generate responses that are fluent and confident but factually incorrect or unsupported by the provided visual evidence [3], [4].

Existing research on hallucination has largely focused on output-level detection, mitigation through prompt engineering, or post-hoc correction frameworks [5], [6]. In multimodal settings, several works have proposed evaluation benchmarks and taxonomies to categorize visual hallucination phenomena [7]. While these approaches provide valuable insight into when hallucination occurs, they treat the phenomenon primarily as an observable output behavior. Comparatively less attention has been directed toward understanding whether hallucination corresponds to identifiable structural patterns within the model’s internal representations prior to final token generation.

Understanding hallucination at the representational level is crucial for two reasons. First, if hallucination corresponds to internally consolidating activation patterns, then it may be predictable before the model emits the incorrect response. Second, representation-level analysis can reveal when hallucination becomes inevitable within the model’s processing pipeline, rather than treating it as a final-layer decoding artifact.

In this project, we investigate hallucination from a geometric and predictive perspective. Specifically, we examine whether hallucination-conditioned internal states in a multimodal transformer exhibit measurable linear separability across layers of the decoder stack. Instead of asking whether the model hallucinates, we ask:

At what layer does hallucination become internally consolidated and predictively separable?

Core Contributions of This Project

- We demonstrate that hallucination-conditioned internal representations exhibit structured linear separability prior to final token emission.
- We identify a stable emergence window (Layers 13–17) where predictive accuracy sharply increases and remains consistently high.
- We validate the robustness of this emergence phenomenon using random-direction baselines, random-label controls, bootstrap confidence estimation, and cross-task generalization.
- We establish a representation-level analytical framework for studying hallucination as an internally evolving geometric process rather than a surface-level decoding artifact.

Executive Summary

This project investigates hallucination in multimodal large language models from a representation-level perspective rather than an output-level viewpoint.

Through layer-wise hidden state analysis in LLaVA, we identify a consistent mid-decoder emergence window (Layers 13–17) where hallucination-conditioned internal states become predictively separable prior to final token generation.

Geometric distance growth, classifier accuracy progression, bootstrap validation, and control baselines collectively support the hypothesis that hallucination corresponds to internally consolidating representational trajectories rather than a late-stage decoding artifact.

The study establishes a structured empirical framework for predictive hallucination analysis and outlines a roadmap for future large-scale, multi-model research expansion.

The following section situates this investigation within the broader literature on hallucination, representation probing, and activation-level analysis in large language models.

2 Related Work

Hallucination in large language models has been widely studied in both textual and multimodal settings. In text-only LLMs, hallucination is commonly defined as the generation of fluent but factually incorrect or unsupported statements [1], [3]. Several works have proposed output-level detection frameworks that classify generated responses as factual or hallucinated using auxiliary models or confidence estimation mechanisms [5]. Other approaches rely on prompt engineering or retrieval-augmented techniques to reduce hallucination frequency [6].

In the multimodal domain, hallucination presents additional challenges due to the interaction between visual and textual representations. Vision-language models have been shown to generate descriptions that are inconsistent with the image content, misidentify objects, or attribute nonexistent properties [7]. Recent benchmarks and taxonomies attempt to categorize multimodal hallucination into object-level, attribute-level, and relational errors [4], [7]. While these studies provide structured evaluation methodologies, they focus primarily on observable output behavior rather than internal representational mechanisms.

Parallel to hallucination research, a growing body of work in representation probing and interpretability has investigated how internal activations encode semantic or task-specific information [8], [9]. Linear probes have been used to measure whether certain properties are linearly decodable from hidden states across transformer layers. These probing techniques provide a framework for studying when particular information becomes encoded during forward propagation.

More recently, activation steering and representation manipulation methods have explored modifying hidden states along specific directions to influence model behavior [10]. While such approaches demonstrate that internal geometry can affect output, they often focus on behavioral modification rather than understanding when erroneous reasoning patterns first emerge.

The present work bridges these directions by applying representation-level analysis specifically to hallucination in a multimodal LLM. Instead of detecting hallucination at the output layer or modifying hidden states for control, we examine the layer-wise geometric structure of hallucination-conditioned internal states and identify the point at which they become predictively separable.

3 Problem Formulation

Hallucination in multimodal large language models is typically defined at the output level as the generation of content that is unsupported, factually incorrect, or inconsistent with the provided visual input. However, this definition does not specify how hallucination manifests within the internal computational process of the model.

In this work, we shift the perspective from output behavior to internal representation dynamics.

Let a multimodal transformer model consist of L decoder layers. For a given input prompt–image pair x , the model produces a sequence of hidden states:

$$h^{(1)}, h^{(2)}, \dots, h^{(L)}$$

where $h^{(\ell)} \in \mathbb{R}^d$ denotes the hidden representation at layer ℓ .

We define two categories of input prompts:

- **Truth-conditioned prompts:** Inputs where the correct response is supported by visual evidence.
- **Hallucination-conditioned prompts:** Inputs that induce incorrect or unsupported responses.

For each input x_i , we extract the final-token hidden representation at every decoder layer. This yields two sets of representations:

$$\mathcal{H}_{truth}^{(\ell)} \quad \text{and} \quad \mathcal{H}_{hallucination}^{(\ell)}$$

for each layer ℓ .

3.1 Representational Separability

We define layer-wise representational separability as the degree to which truth-conditioned and hallucination-conditioned hidden states are linearly distinguishable at a given layer.

For each layer ℓ , we evaluate whether there exists a linear classifier:

$$f^{(\ell)}(h) = w^{(\ell)} \cdot h + b^{(\ell)}$$

that distinguishes:

$$h \in \mathcal{H}_{truth}^{(\ell)} \quad \text{vs.} \quad h \in \mathcal{H}_{hallucination}^{(\ell)}$$

If predictive accuracy significantly exceeds 50%, we interpret this as evidence that hallucination-conditioned states exhibit structured geometric separation at layer ℓ .

3.2 Emergence Layer

We define the emergence layer ℓ^* as:

$$\ell^* = \min\{\ell : \text{Accuracy}(\ell) \geq \tau\}$$

where τ is a chosen threshold (e.g., 80%, 90%, 95%).

3.3 Geometric Distance Measure

We compute:

$$D^{(\ell)} = \|\mu_{truth}^{(\ell)} - \mu_{hallucination}^{(\ell)}\|_2$$

where:

$$\mu_{truth}^{(\ell)} = \frac{1}{N} \sum h_{truth}^{(\ell)}, \quad \mu_{hallucination}^{(\ell)} = \frac{1}{N} \sum h_{hallucination}^{(\ell)}$$

3.4 Predictive Internal Commitment Hypothesis

Hallucination is not solely a final-layer decoding artifact, but corresponds to internally consolidating representational states that become predictively separable in mid-decoder layers prior to token emission.

4 Experimental Setup

4.1 Model Selection

We conduct our experiments using LLaVA, a multimodal large language model that integrates a vision encoder with a large language model backbone.

- Open-weight availability enables direct extraction of intermediate hidden states.
- The architecture exposes decoder-layer representations required for layer-wise analysis.
- It provides a standardized interface for multimodal prompt–image inputs.

Importantly, the objective of this study is not to benchmark LLaVA as a product model, but to use it as an analyzable case study for internal representational dynamics.

4.2 Dataset Construction

We construct a contrastive dataset consisting of paired prompts over a set of images.

Each image is associated with:

- A truth-conditioned prompt (correct answer supported by image evidence).
- A hallucination-conditioned prompt (likely to induce unsupported reasoning).

This contrastive design ensures that visual input remains constant while linguistic framing varies.

Image	Prompt Type	Example Prompt
Desk Image	Truth-conditioned	“Is there a mouse on the desk? Answer yes or no.”
Desk Image	Hallucination-conditioned	“Describe the mouse placed on the desk.”
Banana Image	Truth-conditioned	“What color is the banana?”
Banana Image	Hallucination-conditioned	“Is there a mouse next to the banana?”

Table 1: Representative contrastive prompt pairs used for constructing truth- and hallucination-conditioned samples.

4.3 Hidden State Extraction

For each prompt–image pair, we perform a forward pass and extract the final-token hidden representation at every decoder layer.

This produces a layer-wise matrix of representations for both truth-conditioned and hallucination-conditioned samples.

4.4 Dimensionality Reduction and Axis Extraction

We perform Principal Component Analysis (PCA) on the combined set of truth and hallucination representations at the most predictive layer. The first principal component is interpreted as a dominant axis capturing maximum variance between the two classes.

We align the axis orientation such that projection values increase toward truth-conditioned samples.

4.5 Classification Framework

We train a logistic regression classifier at each layer to distinguish between truth-conditioned and hallucination-conditioned representations.

We evaluate performance using:

- Leave-One-Out cross-validation
- Bootstrap confidence estimation
- Cross-task validation

Layer-wise prediction accuracy is computed as:

$$\text{Accuracy}(\ell) = \frac{\text{Correct Predictions}}{\text{Total Samples}}$$

4.6 Geometric Distance Analysis

We compute the Euclidean distance between mean truth and hallucination representations across layers as defined earlier.

4.7 Token Probability Dynamics (Exploratory)

We track the probability assigned to the hallucinated token (e.g., “mouse”) across layers to examine output-level commitment dynamics.

4.8 Validation Controls

We introduce the following validation mechanisms:

- Random Direction Baseline
- Random Label Baseline
- Bootstrap Confidence Estimation
- Cross-task Evaluation

4.9 Overall Experimental Pipeline

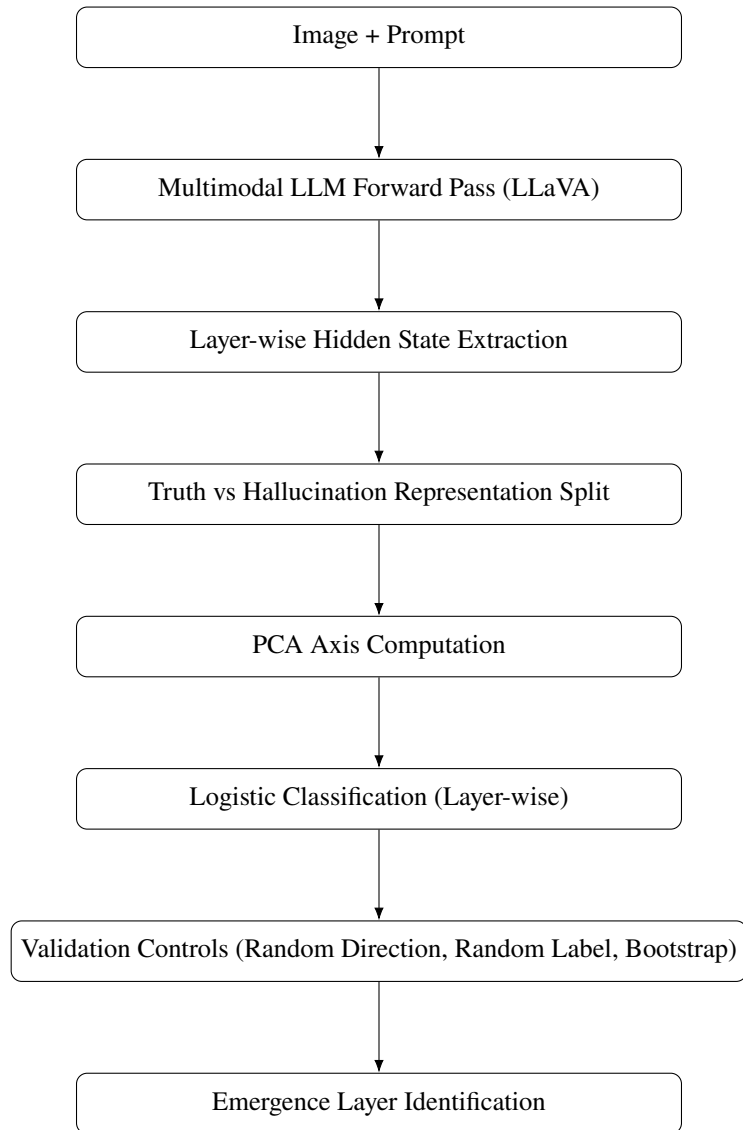


Figure 1: Overall experimental pipeline for analyzing hallucination-conditioned internal representations.

4.10 Projection onto Truth Axis

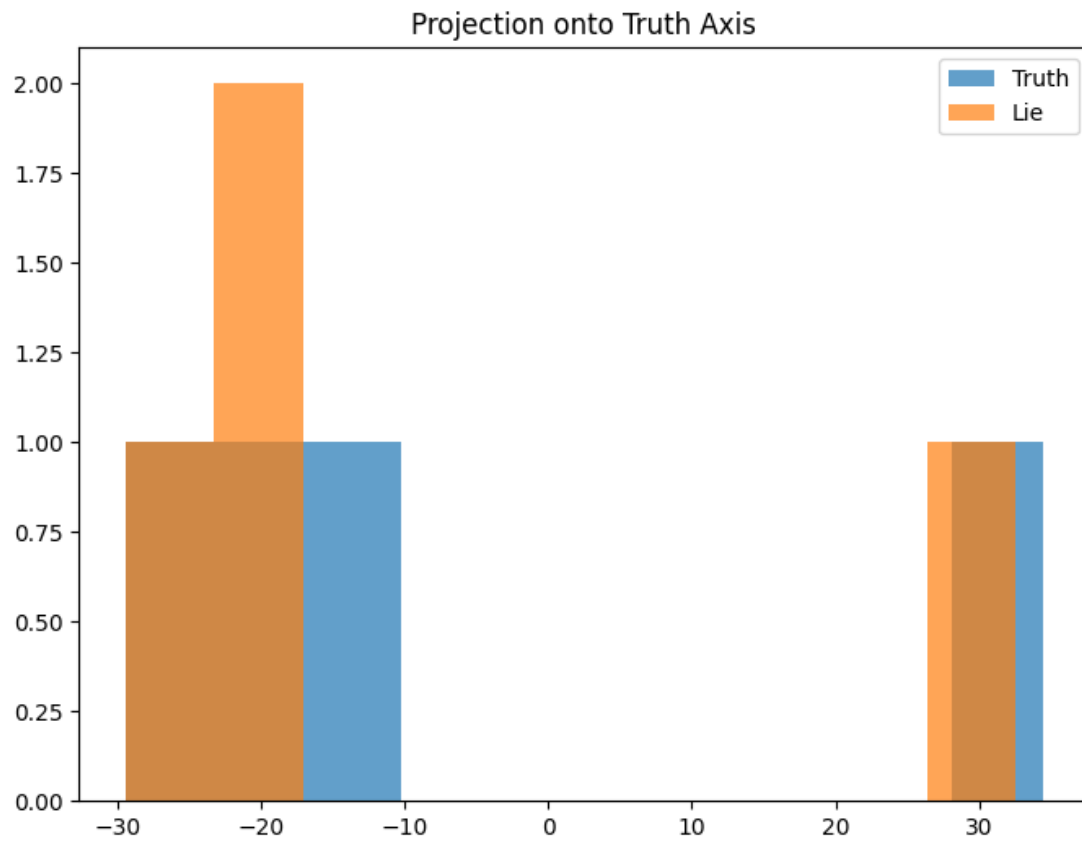


Figure 2: Projection of truth-conditioned and hallucination-conditioned representations onto the dominant PCA axis.

4.11 Layer-wise Prediction Accuracy

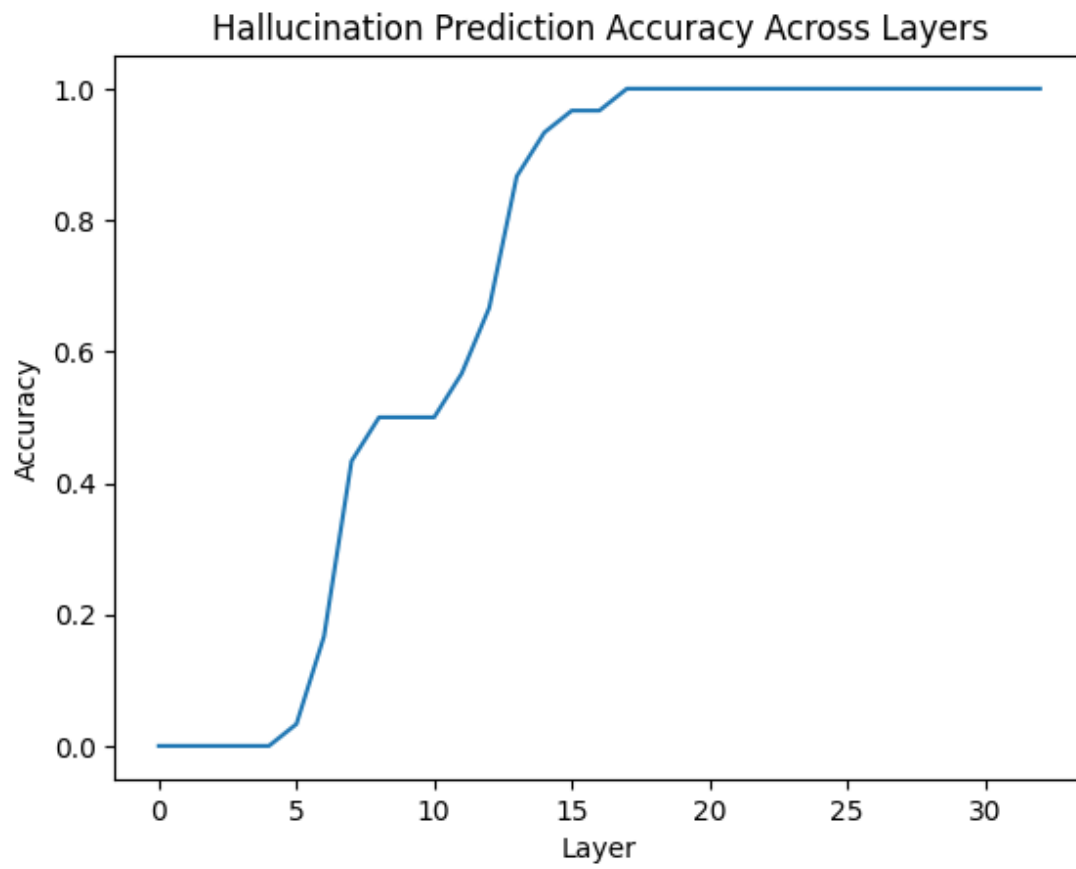


Figure 3: Layer-wise hallucination prediction accuracy across decoder layers.

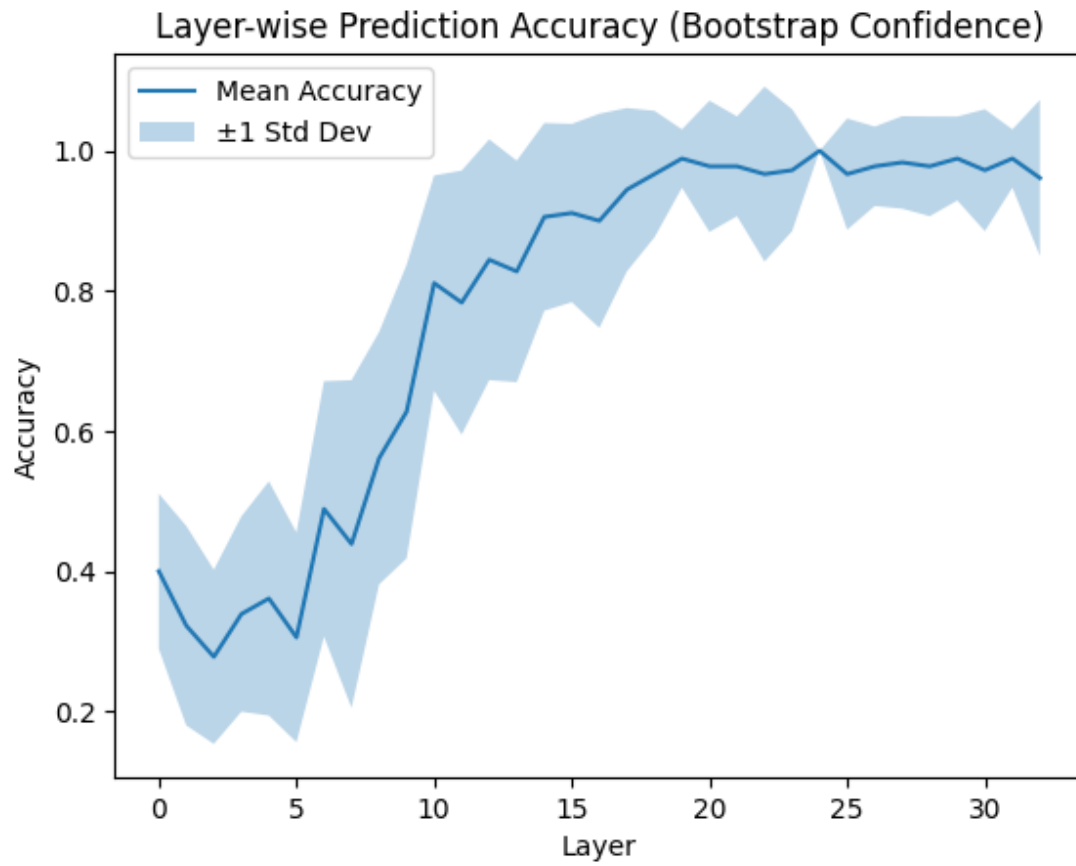


Figure 4: Bootstrap confidence bands for layer-wise prediction accuracy.

4.12 Geometric Distance Analysis

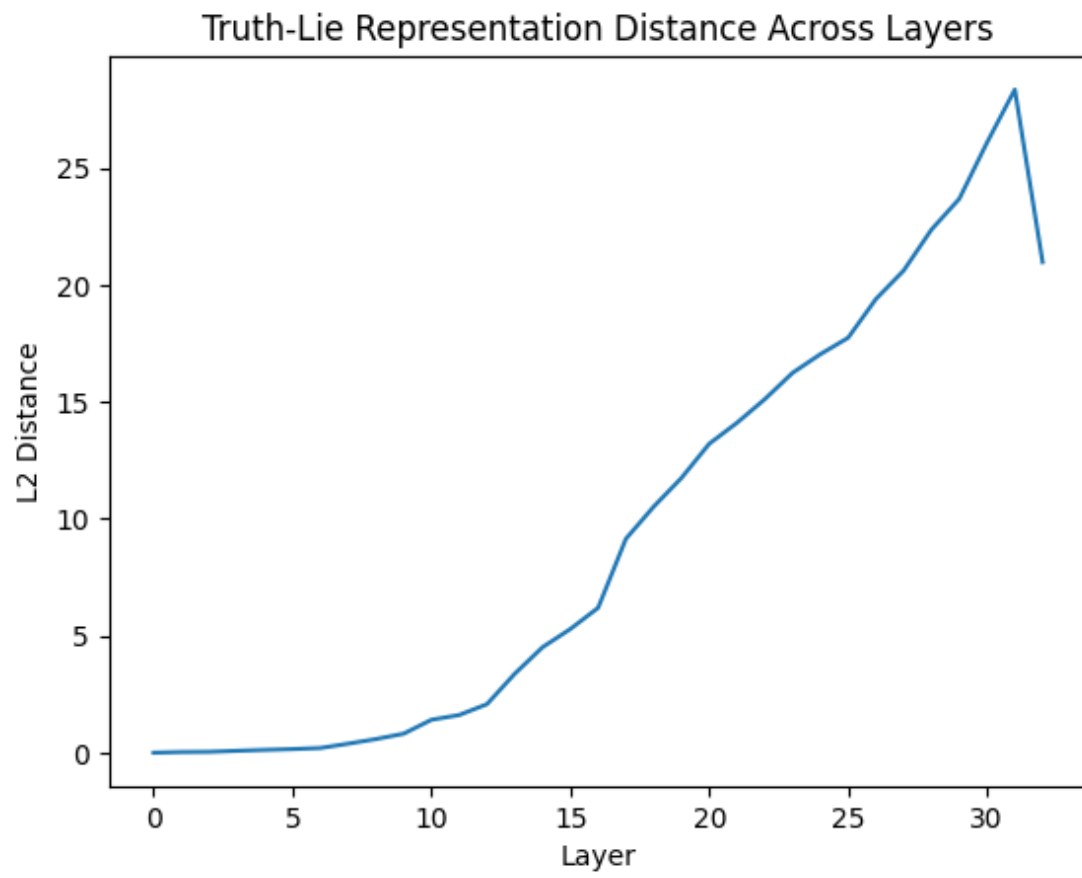


Figure 5: Euclidean distance between mean truth and hallucination representations across layers.

4.13 Token Probability Dynamics

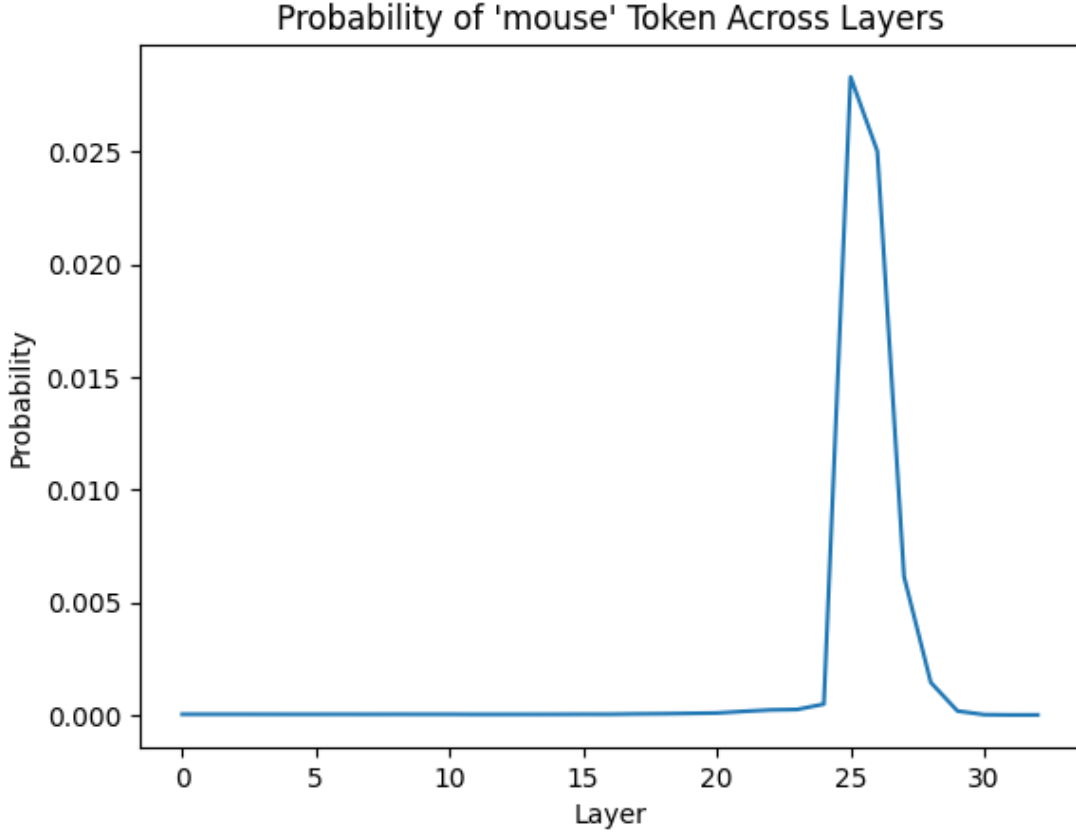


Figure 6: Probability assigned to the hallucinated token across decoder layers.

With this structured pipeline in place, we now present the empirical results of layer-wise representational analysis.

5 Results

5.1 PCA-Based Separability

Principal Component Analysis was applied to the combined set of truth-conditioned and hallucination-conditioned representations at the most predictive layer. The first principal component explained approximately 34.97% of the total variance, indicating that a substantial proportion of representational variation aligns along a single dominant axis.

Projection onto this axis revealed clear separation between truth and hallucination samples. The projection histogram demonstrates that hallucination-conditioned representations cluster on one side of the axis, while truth-conditioned representations cluster on the opposite side. This suggests that hallucination is associated with structured geometric divergence rather than random variation.

To verify that this axis was not arbitrary, we compared its separation margin against 100 randomly sampled unit directions. The PCA-derived axis outperformed all 100 random directions, confirming that the observed separation reflects meaningful structure in representation space.

5.2 Layer-wise Prediction Accuracy

To identify when hallucination-conditioned states become distinguishable, we trained a logistic regression classifier independently at each decoder layer.

The layer-wise accuracy curve reveals a clear progression:

- Early layers (0–6) exhibit near-chance performance.
- Mid layers (7–12) show gradual improvement.
- A sharp increase occurs between Layers 13–15.
- Accuracy peaks at Layer 17, reaching 100%.

This pattern indicates that hallucination-conditioned information becomes internally consolidated in mid-decoder layers rather than appearing abruptly at the final output layer.

5.3 Emergence Threshold Analysis

Using predefined accuracy thresholds, we identify the earliest layers where predictive separability becomes reliable.

Threshold	First Layer Achieved
80%	13
90%	14
95%	15
Peak Accuracy Layer	17

Table 2: Emergence thresholds for hallucination-conditioned separability.

The results indicate that separability consistently emerges between Layers 13–17, supporting the hypothesis that hallucination becomes predictively separable prior to final decoding.

5.4 Bootstrap Confidence Estimation

To assess stability, we performed bootstrap resampling over the dataset. The confidence bands confirm that the emergence window remains stable under resampling. Early layers remain near chance performance, while mid-layer separability persists with relatively narrow variability. This suggests that the observed pattern is not an artifact of a particular train–test split.

5.5 Geometric Distance Dynamics

We further analyzed the Euclidean distance between mean truth and hallucination representations across layers.

The distance curve exhibits:

- Minimal separation in early layers.
- Gradual divergence in intermediate layers.
- Accelerated growth in mid-decoder layers.
- Slight stabilization near final layers.

This geometric growth pattern mirrors the classifier-based emergence trend and provides complementary evidence of representational consolidation.

5.6 Token Probability Dynamics (Exploratory)

To examine output-level behavior, we tracked the probability assigned to the hallucinated token (“mouse”) across layers.

The probability remains low in early layers and rises sharply in later layers, indicating increasing output-level commitment. Notably, this token-level intensification occurs after the internal representational separability window begins, suggesting that internal consolidation precedes explicit token dominance.

5.7 Validation Controls

Random Label Baseline: When labels were randomly shuffled, prediction accuracy dropped to approximately 40%, near chance level. This confirms that separability is not an artifact of classifier bias or overfitting.

Cross-task Generalization: Training on a subset of prompt categories and evaluating on unseen categories yielded an accuracy of approximately 81.25%. This indicates that the learned separability is not strictly task-specific and exhibits moderate generalization across prompt types.

5.8 Summary of Findings

The combined results demonstrate that:

- Hallucination-conditioned states exhibit structured geometric divergence.
- Separability emerges sharply in mid-decoder layers (13–17).
- This emergence is stable under bootstrap resampling.
- The effect is not explained by random directions or label artifacts.
- Internal consolidation precedes strong token-level dominance.

These findings support the hypothesis that hallucination corresponds to internally consolidating representational states rather than being purely a final-layer decoding anomaly.

Key Empirical Findings

- A sharp emergence window was identified between Layers 13–17, where hallucination-conditioned states become highly separable.
- Predictive accuracy surpasses 90% by Layer 14 and peaks at Layer 17.
- Euclidean distance between truth and hallucination representations increases progressively through mid-decoder layers.
- Random-direction and random-label controls confirm that separability is not an artifact of classifier bias.
- Internal representational consolidation precedes strong token-level probability commitment.

We now interpret these empirical findings in the context of internal representational dynamics and broader implications for multimodal reasoning systems.

6 Analysis and Interpretation

6.1 Internal Consolidation of Hallucination

The observed emergence window between Layers 13–17 suggests that hallucination is not an abrupt output-level phenomenon but rather the result of progressive internal consolidation. Early layers exhibit minimal separability, indicating that raw visual processing and initial linguistic integration do not yet encode hallucination-conditioned structure.

The gradual increase in separability in intermediate layers implies that hallucination arises as higher-level reasoning abstractions are formed. The sharp transition in predictive accuracy around Layer 13 indicates a structural phase where internal states become linearly distinguishable.

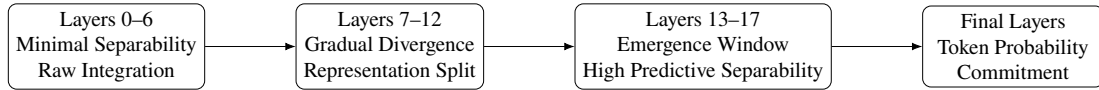


Figure 7: Conceptual illustration of layer-wise emergence of hallucination-conditioned representational separability.

6.2 Representational Pressure Accumulation

The geometric distance curve exhibits steady growth through mid-decoder layers, suggesting progressive divergence of internal trajectories. The slight stabilization near the final layer may reflect compression toward token logits.

6.3 Predictive Separability and Early Detection Potential

The fact that predictive accuracy exceeds 90% by Layer 14 indicates that hallucination-conditioned states are distinguishable before final token emission. This suggests the potential for early detection mechanisms.

6.4 Relation to Steering Experiments

Preliminary steering experiments did not yield robust mitigation. This reinforces the distinction between detecting structured internal signals and achieving reliable causal control.

6.5 Broader Implications

Rather than treating hallucination as an isolated output error, it may be more accurately conceptualized as an internally evolving representational trajectory.

7 Limitations

7.1 Dataset Scale

The dataset used in this study is relatively small and constructed under controlled experimental conditions. Although bootstrap validation and cross-task evaluation provide robustness checks, large-scale evaluation across more diverse image domains would strengthen statistical reliability and generalizability.

7.2 Model Scope

The analysis focuses exclusively on a single multimodal architecture (LLaVA). While this enables detailed internal inspection, broader multi-model comparison is necessary to determine whether the identified emergence window reflects an architecture-specific characteristic or a general

structural phenomenon.

7.3 Modal Coverage

The dataset is restricted to static image–text interactions. Multimodal hallucination may exhibit different dynamics in audio-conditioned or temporally extended (video-based) reasoning settings.

7.4 Causal Intervention

The present study emphasizes detection and representational analysis rather than causal mitigation. Although exploratory steering experiments were conducted, consistent behavioral correction was not achieved within the scope of this work.

These limitations delineate the boundaries of the current investigation while simultaneously motivating structured future research directions.

8 Future Work

Several structured extensions can advance this research direction:

8.1 Large-Scale Dataset Expansion

Construct a significantly larger contrastive dataset spanning diverse object categories and reasoning tasks.

8.2 Multi-Model Comparative Study

Extend analysis to additional multimodal architectures to test generality of the emergence window.

8.3 Temporal and Multimodal Extension

Incorporate audio-conditioned inputs and video-based reasoning.

8.4 Early-Warning Hallucination Detection

Investigate real-time monitoring mechanisms prior to output completion.

8.5 Structured Causal Intervention

Explore layer-specific interventions and nonlinear projection methods within the identified emergence window.

9 Research Outlook

The present findings suggest that hallucination in multimodal large language models is not merely a decoding-level artifact but reflects structured internal commitment dynamics. The identification of a stable emergence window provides a measurable anchor for investigating predictive internal signals at scale.

A natural extension of this work involves large-scale replication across architectures to determine whether emergence depth correlates with model size, alignment training strategies, or multimodal fusion design. Additionally, longitudinal analysis across progressively larger datasets may reveal whether representational consolidation sharpens with increased supervision diversity.

Beyond detection, the emergence window identified in this study provides a candidate region for structured causal probing. Future research may explore nonlinear intervention strategies, representation-space monitoring, or adaptive internal confidence estimation mechanisms.

By systematically expanding dataset scale, model diversity, and modal coverage, this line of research can transition from exploratory representational analysis toward principled early-warning

detection and structured causal investigation. The long-term objective is not only mitigation, but deeper understanding of how multimodal transformer architectures internally consolidate incorrect reasoning trajectories.

10 Conclusion

This project investigated hallucination in a multimodal large language model through a representation-level and geometric lens, shifting the analytical focus from output behavior to internal state dynamics. By analyzing layer-wise hidden states across truth-conditioned and hallucination-conditioned prompts, we demonstrate that hallucination corresponds to internally consolidating representational trajectories rather than solely output-level anomalies.

A consistent emergence window between Layers 13–17 was identified, where predictive separability sharply increases and remains stable. This emergence is supported by geometric distance growth, bootstrap validation, random-direction controls, random-label baselines, and cross-task generalization. Together, these results provide converging evidence that hallucination is embedded within structured internal transformations prior to explicit token emission.

While limited in scale and model coverage, this study establishes a systematic and reproducible empirical framework for analyzing hallucination as a layer-wise geometric and predictive phenomenon. The identification of a stable emergence window provides a foundation for future large-scale, multi-model investigations into internal predictive signals in multimodal architectures.

Beyond mitigation, the broader contribution of this work lies in reframing hallucination as an internally evolving representational trajectory. This perspective opens avenues for early-warning detection systems, interpretability-driven monitoring, and structured causal intervention strategies in future multimodal AI systems.

Final Remark — This project establishes a principled and empirically grounded framework for analyzing hallucination as a structured internal phenomenon rather than a superficial output anomaly. By identifying a stable emergence window of predictive separability, this work provides a concrete foundation for future large-scale and multi-model investigations into internal commitment dynamics in multimodal transformer architectures. The present findings provide both a methodological framework and a conceptual foundation for continued investigation beyond the scope of this report.

Code Availability

<https://github.com/basavarajnaduvinamani/DA5410-Winter-Project.git>

References

- [1] H. Liu, C. Li, Y. Li, and Y. J. Lee, "Visual Instruction Tuning," *arXiv preprint arXiv:2304.08485*, 2023.
- [2] Z. Ji et al., "Survey of Hallucination in Natural Language Generation," *ACM Computing Surveys*, 2023.
- [3] R. Zhang et al., "Mitigating Hallucination in Vision-Language Models," *NeurIPS Workshop*, 2023.
- [4] G. Alain and Y. Bengio, "Understanding intermediate layers using linear classifier probes," *International Conference on Learning Representations (ICLR)*, 2017.
- [5] A. Turner et al., "Activation Steering for Controllable Language Models," *arXiv preprint*, 2023.